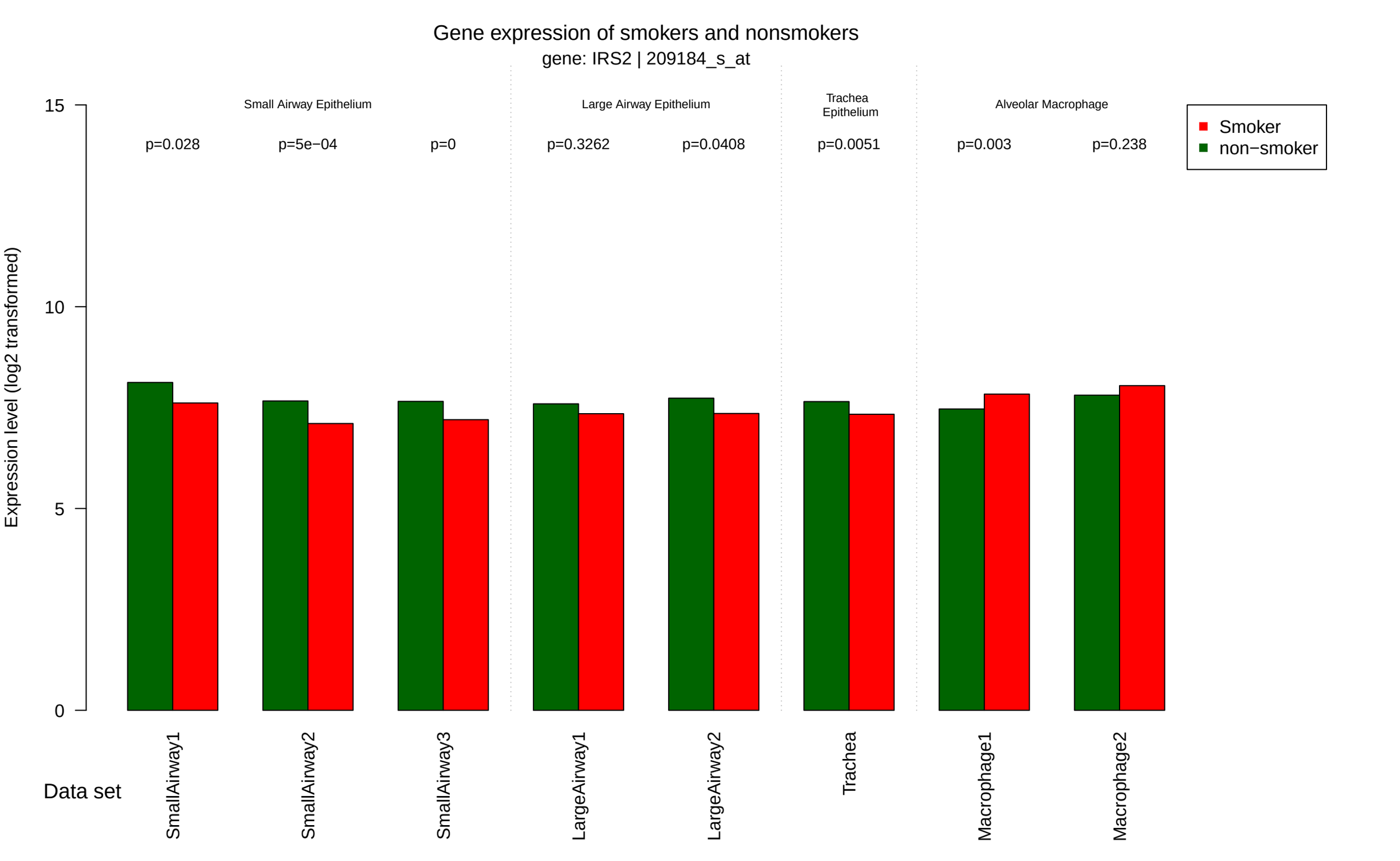
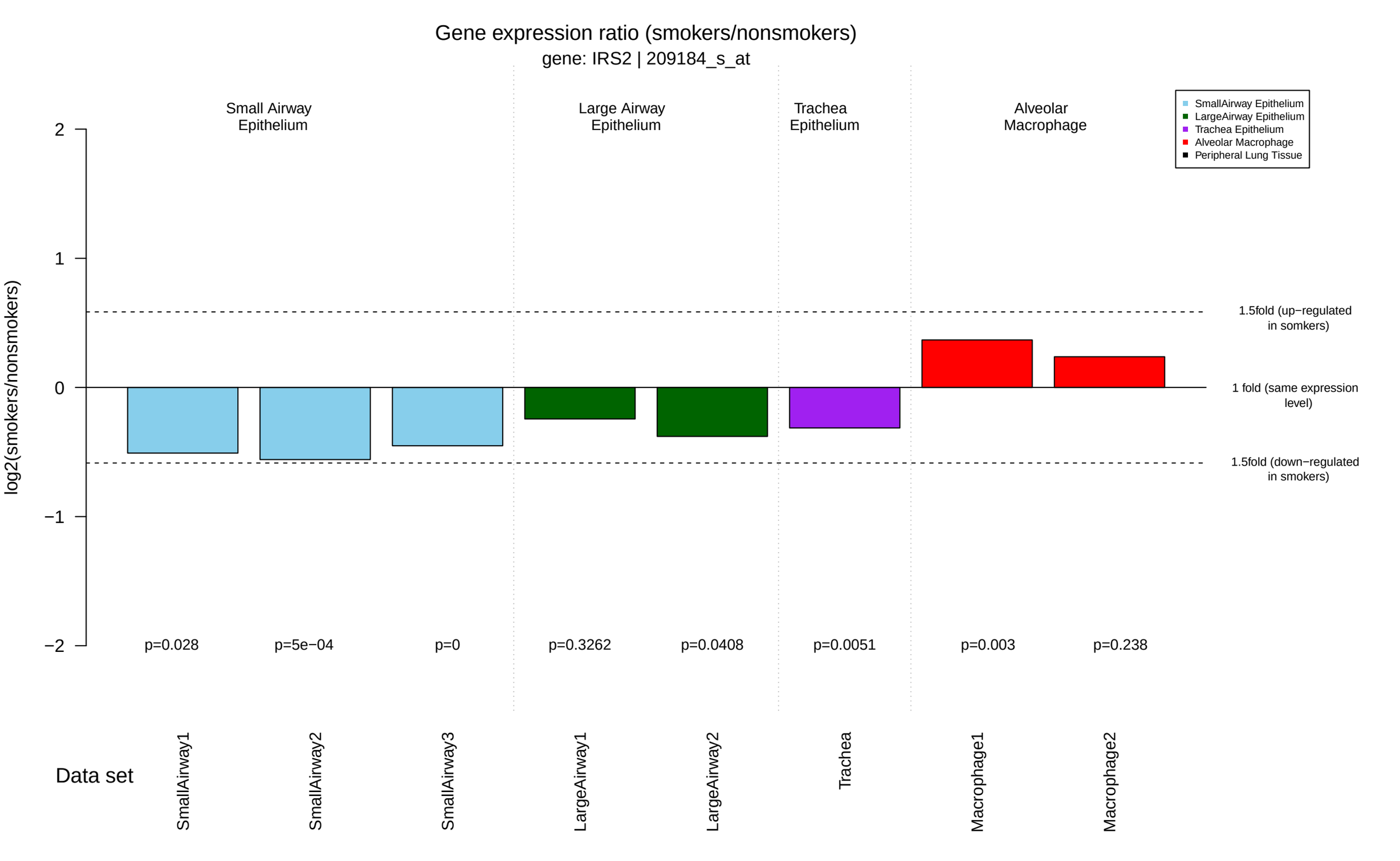
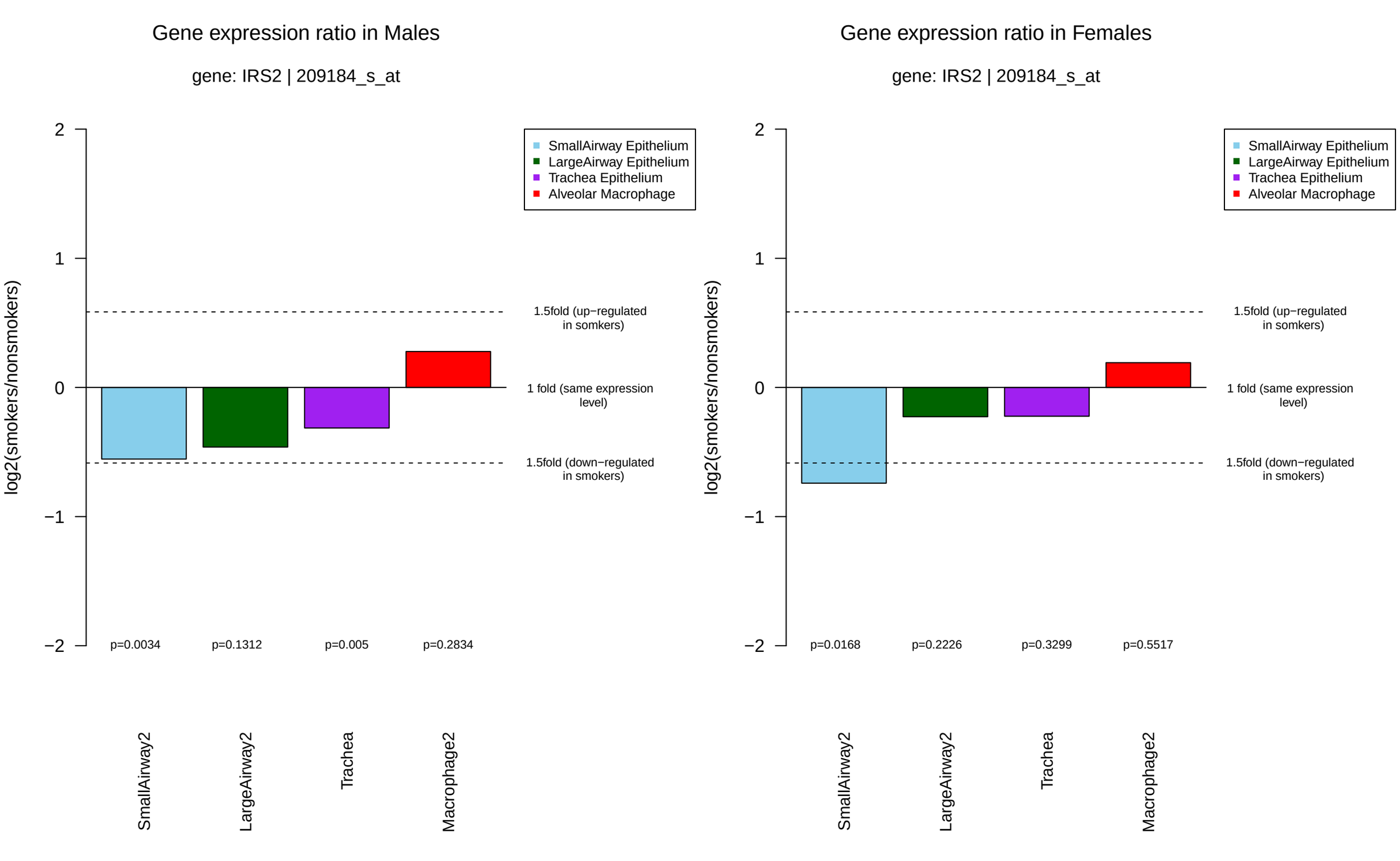
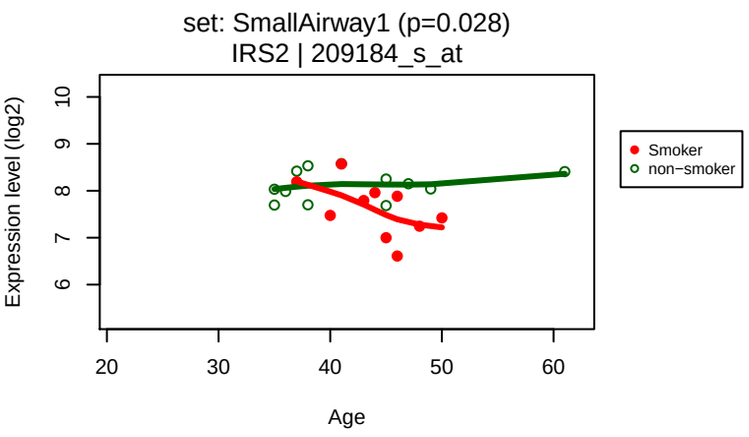
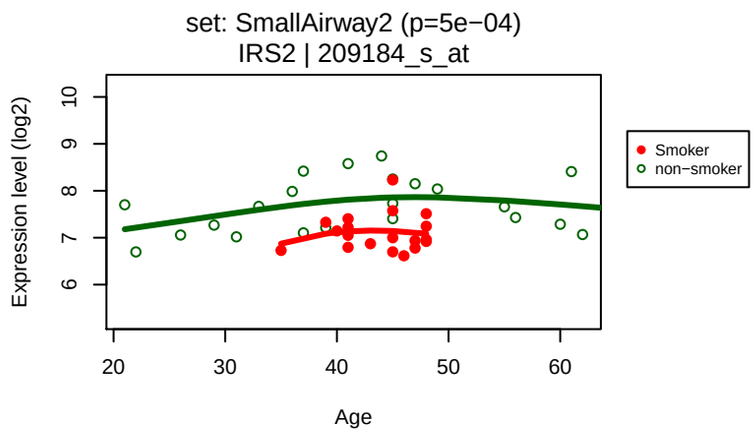
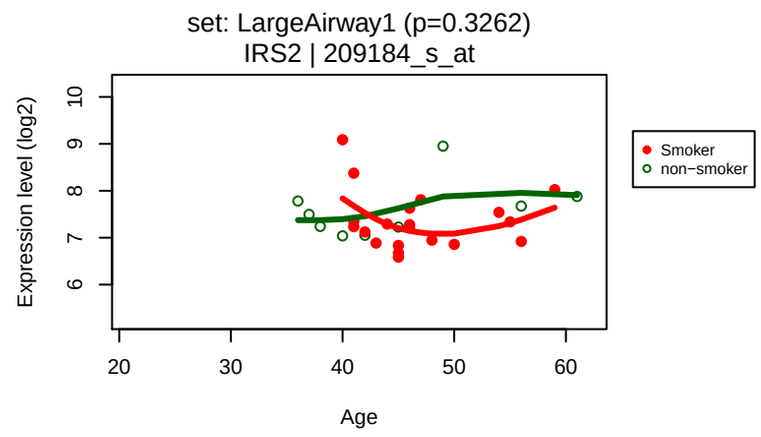

Gene expression of smokers and nonsmokers
gene: IRS2 | 209184_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.028
p=5e−04
p=0
p=0.3262
p=0.0408
p=0.0051
p=0.003
p=0.238
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

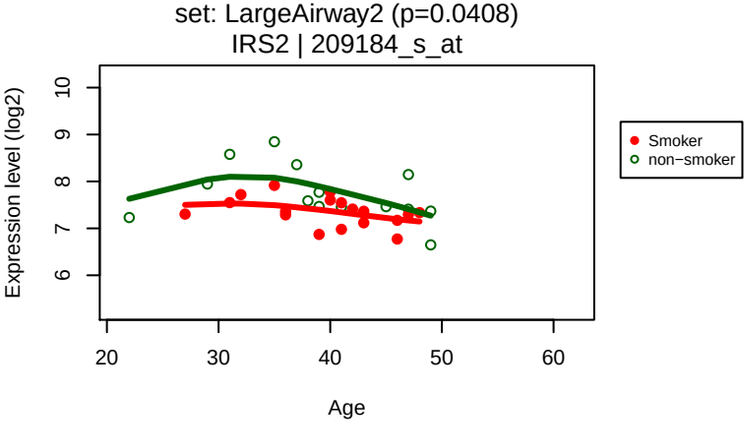

Gene expression ratio (smokers/nonsmokers)
gene: IRS2 | 209184_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.028
p=5e−04
p=0
p=0.3262
p=0.0408
p=0.0051
p=0.003
p=0.238
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

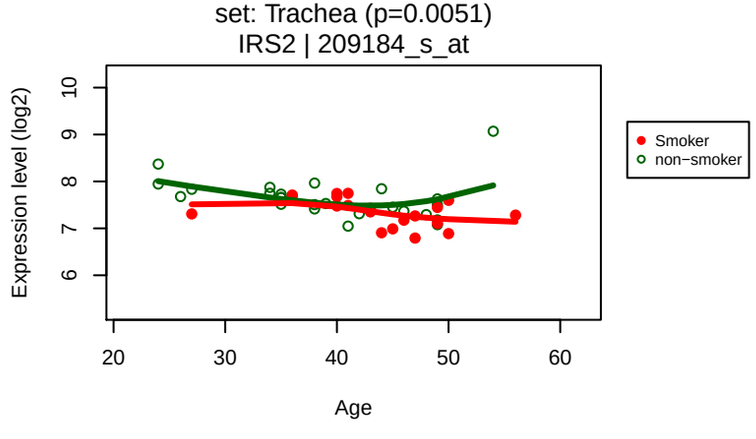

Gene expression ratio in Males
gene: IRS2 | 209184_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0034
p=0.1312
p=0.005
p=0.2834
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRS2 | 209184_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0168
p=0.2226
p=0.3299
p=0.5517
SmallAirway2
LargeAirway2
Trachea
Macrophage2

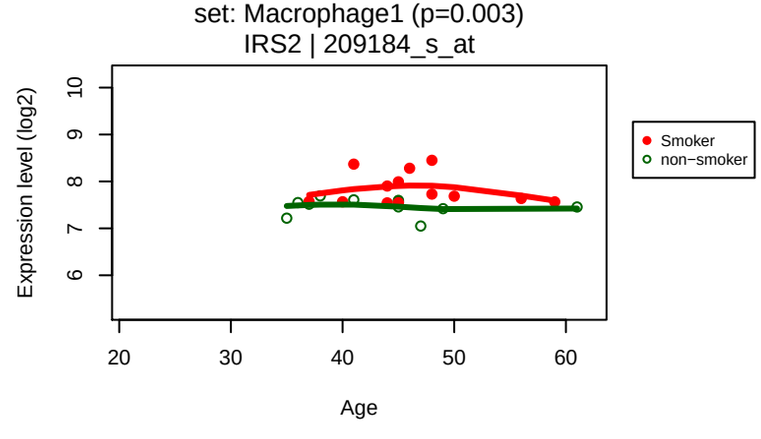

set: SmallAirway1 (p=0.028)
IRS2 | 209184_s_at
Expression level (log2)
Age
Smoker
non−smoker

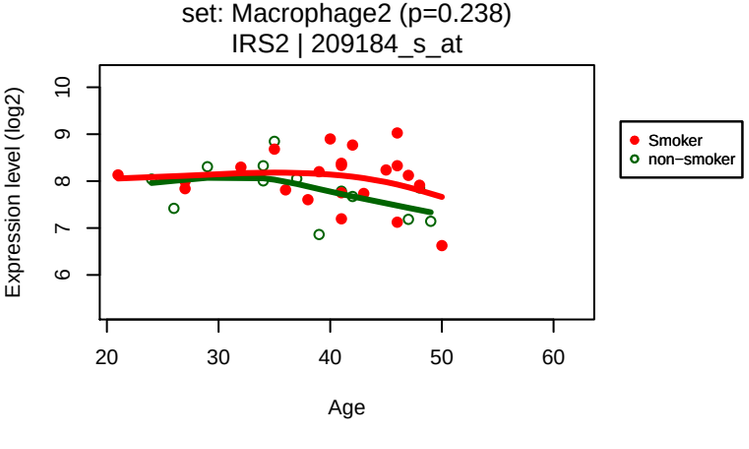

set: SmallAirway2 (p=5e−04)
IRS2 | 209184_s_at
Expression level (log2)
Age
Smoker
non−smoker

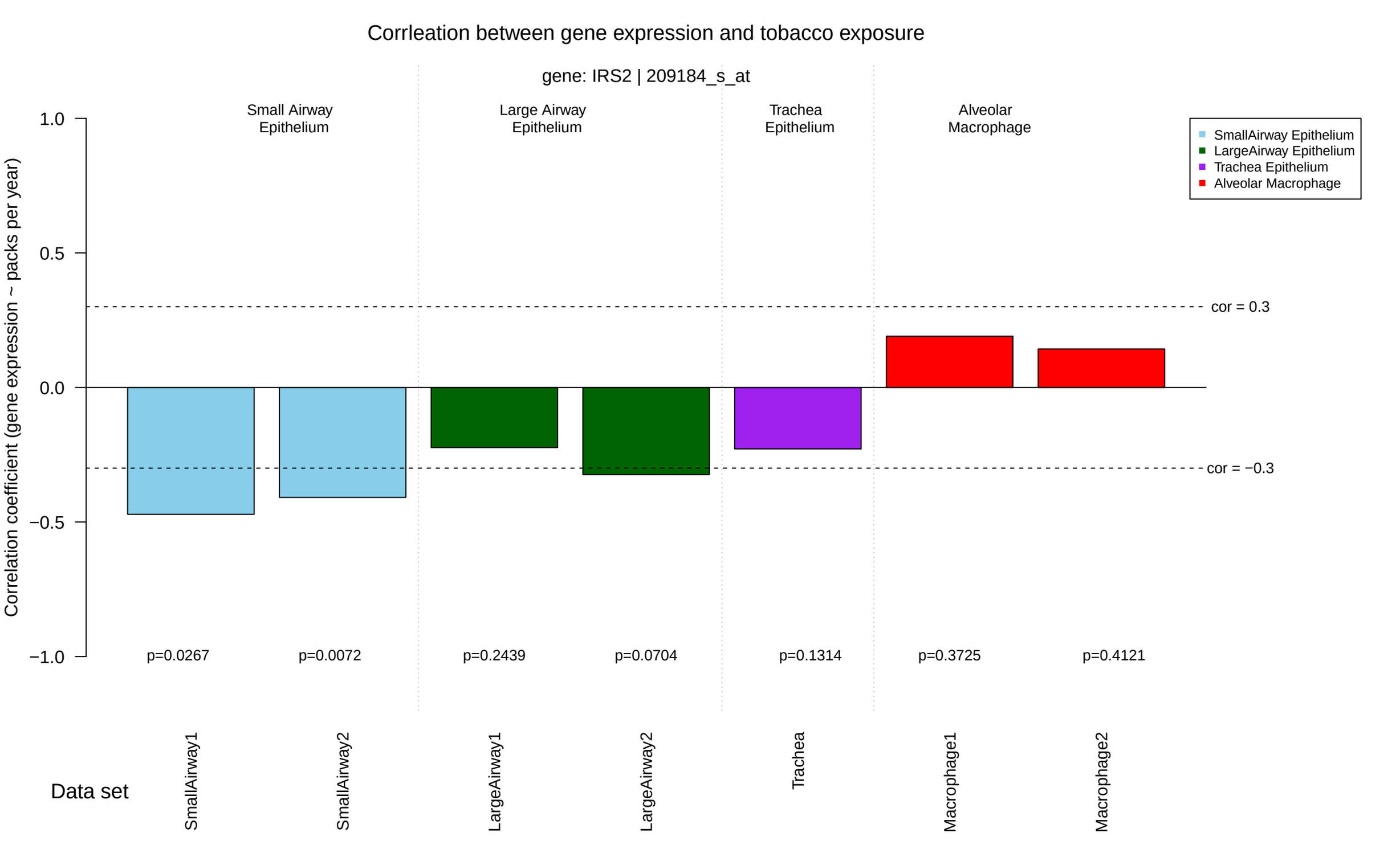

set: LargeAirway1 (p=0.3262)
IRS2 | 209184_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0408)
IRS2 | 209184_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0051)
IRS2 | 209184_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.003)
IRS2 | 209184_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.238)
IRS2 | 209184_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRS2 | 209184_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0267
p=0.0072
p=0.2439
p=0.0704
p=0.1314
p=0.3725
p=0.4121
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2